

Stage 2: Visual information and communication context as modulators of interpersonal coordination in face-to-face and videoconference-based interactions[☆]

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ABSTRACT

Interpersonal coordination of body movement—or similarity in patterning and timing of body movement between interaction partners—is well documented in face-to-face (FTF) conversation. Here, we investigated the degree to which interpersonal coordination is impacted by the amount of visual information available *and* the type of interaction conversation partners are having. To do so within a naturalistic context, we took advantage of the increased familiarity with videoconferencing (VC) platforms and with limited visual information in FTF conversation due to the COVID-19 pandemic. Pairs of participants communicated in one of three ways: FTF in a laboratory setting while socially distanced and wearing face masks; VC in a laboratory setting with a view of one another's full movements; or VC in a remote setting with a view of one another's face and shoulders. Each pair held three conversations: affiliative, argumentative, and cooperative task-based. We quantified interpersonal coordination as the relationship between the two participants' overall body movement using nonlinear time series analyses. Coordination changed as a function of the contextual constraints, and these constraints interacted with coordination patterns to affect subjective conversation outcomes. Importantly, we found patterns of results that were distinct from previous research; we hypothesize that these differences may be due to changes in the broader social context from COVID-19. Taken together, our results are consistent with a dynamical systems view of social phenomena, with interpersonal coordination emerging from the interaction between components, constraints, and history of the system.

1. Visual information and communication context as modulators of interpersonal coordination in face-to-face and videoconference-based interactions

Interpersonal coordination or interpersonal synchrony can be defined as the similarity in patterning of body movement between interacting individuals over time (Shockley et al., 2009). Studies of interpersonal coordination have increasingly often focused on identifying the conditions or contexts under which the phenomenon emerges and changes. Here, we build on previous research to investigate the degree to which interpersonal coordination between two people is shaped by the visual information available about a conversation partner and type of interaction that the conversation partners are having. Critically, we

capitalized on the “new normal” (in much of the world) induced by the COVID-19 pandemic to incorporate a “new naturalism” into our study.

1.1. Existing comparisons of videoconferencing and face-to-face communication

Prior work on the effects of engaging remotely over videoconferencing (VC) platforms compared to face-to-face (FTF) communication have been mostly centered in the context of work relationships and health. While researchers consistently find that workers' feelings about engaging through VC are more negative than FTF (Blacksmith et al., 2016; Credé & Sniezek, 2003; Denstadli et al., 2011), findings about the impact of communication medium on effectiveness are mixed (Baltes

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et al., 2002; Credé & Sniezek, 2003; O'Neill et al., 2016). This discrepancy has also been observed between research using experimental and observational methods, with experimental work finding negative effects of VC not found in field work observations (Purvanova, 2014).

Similar patterns have emerged from studies of healthcare delivery. Patients in different contexts have reported negative feelings when having to interact with providers over VC rather than FTF (e.g., Barrett & Murphy, 2020; Gordon et al., 2020; Norwood et al., 2018). By contrast, when considering telehealth appointments without comparing them to FTF interactions, research has found positive outcomes in treating various disorders (DelliFraine & Dansky, 2008).

Across both healthcare and industrial-organizational settings, therefore, previous work suggests distinctive *feelings* toward VC versus FTF, though the evidence for an impact on interaction *outcomes* is mixed. However, most of this research has focused its attention on finding *whether* there are differences between interactions depending on the interaction medium and not *why* these differences might be occurring. The current study builds on this existing work by evaluating some possible drivers of these effects.

The COVID-19 pandemic of 2020 changed our lives dramatically as society adapted to the new threat. One of the most visible changes was in our modes of communication: While we collaborated and socialized FTF before, the pandemic forced us to engage remotely over VC platforms or to meet FTF with new public health constraints like mask wearing and social distancing. These changes highlighted the importance of understanding the impacts of communication medium on interaction. The current research leveraged these changes to provide insights into the reported feelings of disconnection due to pandemic abatement measures (deployed to combat COVID-19 and perhaps re-deployed to combat future pandemics) and to contribute to the development of theories of interpersonal coordination.

1.2. Interpersonal coordination

In FTF interactions, much of our communication involves and is affected by the movements of our bodies, from large-scale body movements like gesturing to smaller movements like postural sway (Dale et al., 2013; Shockley et al., 2009). Previous research has found that people's body movements synchronize during conversation (Condon & Ogston, 1966; Nagaoka & Komori, 2008; Ramseyer & Tschacher, 2008), although people are often unaware of it (Lakin & Chartrand, 2003). However, interpersonal coordination is not universal: It is deeply tied to contextual and interpersonal factors. The presence and amount of interpersonal bodily coordination differs according to the type of conversation that two people are having (Paxton & Dale, 2013a, 2017), and interpersonal coordination both reflects and affects interacting individuals' feelings toward one another (Fowler et al., 2008; Lakin et al., 2003; Marsh et al., 2009; Miles et al., 2009; Schmidt & Richardson, 2008). Additionally, the amount of interpersonal coordination modulates the effectiveness of cooperative and problem-solving interactions (Allsop et al., 2016; Miles et al., 2017).

Given the important role that coordination plays in FTF interaction, it is imperative to better understand how these dynamics might be affected by interaction constraints like those brought on by pandemic abatement measures. At a basic level, abatement measures affected the amount of visual information available to each participant in the conversation. During many pre-pandemic FTF conversations, people would have been able to see all or most of one another's body and would be separated by a distance of a few feet (Sussman & Rosenfeld, 1982). By contrast, in FTF interactions during the pandemic, most people wore face masks and stood six feet apart, and during VC, most people only show their shoulders and heads. In both cases, these changes reduce the amount and detail of visual information available. Previous research has shown that the amount of visual access to a communication partner impacts interpersonal coordination during both social and non-social tasks (Richardson et al., 2007; Shockley et al., 2003). The changes in

the amount of visual information across these communication media, then, may deeply impact the emergence of interpersonal coordination.

Importantly, the constraints imposed by pandemic abatement measures affected more than just the *amount* of information available to conversants: In some cases, these measures also impacted the *kind* of information that is available (or not) for communication partners—such as covering much of a partner's facial expressions when wearing masks. Such constraints could also affect the level of interpersonal coordination between participants. However, the lack of previous research in this area limits our ability to predict the impact that these differences may have.

1.3. Virtual interactions

Intuitively, most people feel that interactions through VC platforms are different from those taking place FTF. Many people will think immediately of VC's disruptions to speech. For example, a lagging audio signal can disrupt the flow of communication, and only one speaker can use the audio stream at once, restricting natural backchanneling and interjections (e.g., Schegloff, 1982).

However, VC also substantially impacts *visual* information. People interacting over VC often limit their video frame to include only their head and shoulders, restricting the amount of visual information available to their conversation partners. From the limited research to date on this issue, interactions through audio-video links and audio-only are less efficient, reinforcing the importance of a shared physical space to hands-on collaboration (Fussell et al., 2000; Kraut et al., 1996). Additionally, Schmitt et al. (2018) found that low video quality while collaborating on a hands-on task led to significant changes in movement and speech patterns. These findings resonate with the importance of nonverbal communication to FTF interactions that we reviewed above, but between the relatively limited existing work and the rapidly improving VC quality, additional work is needed.

Despite the link between nonverbal dynamics and rapport, no research to date (to our knowledge) has directly investigated the consequences that interacting through VC may have on efficiency of communication and feelings of connectedness. More importantly, we lack an understanding of how the differences in the *amount*, *quality*, and *type* of nonverbal information in VC affect various kinds of interaction (e.g., collaborative interactions vs. rapport-building ones).

However, we can find some inspiration from related work about how virtual reality (VR) affects human movement in different situations. Importantly, this area of research has shown that people respond differently to videographically generated versus physically generated optic flow displacement (Stoffregen et al., 2004) and that participants coordinate their bodily movements *differently* when interacting in a VR setting than in the real world (Soczawa-Stronczyk & Bocian, 2020; Tolston et al., 2014; Varlet et al., 2013). This last finding is important: We might have expected coordination to decrease in a VR environment, given that the created environment is not as rich as a shared physical environment. However, Tolston et al. (2014) found higher levels of coordination in VR than FTF interactions, which “show[s] the sensitivity of interpersonal coordination to visual constraints” (p. 164). As such, we expect that we may find similar impacts of visual constraints on interpersonal coordination during VC as compared to FTF.

1.4. The current research

Here, we aimed to identify some of the basic processes underlying differences in face-to-face (FTF) and videoconferencing (VC) conversations. At the time of our Stage 1 Registered Report (Romero & Paxton, 2021), this was the first study (to our knowledge) to directly test how VC (as one communication medium) affects interpersonal coordination and satisfaction with the conversation, as compared to FTF (as another communication medium). Building from previous research in FTF interactions, we identified the amount of visual information as one key variable that could impact interaction dynamics and, through that,

interaction outcome. Critically, the present work took advantage of pandemic mitigation recommendations in the United States (i.e., masking and social distancing) to limit visual information in FTF interactions in a way that was relatively naturalistic during the pandemic. To investigate the degree to which social context interacts with communication media, we further compared dynamics and outcomes of different types of conversations across communication media.

To ascertain the impact of visual information, we asked each pair of participants to communicate in one of three ways: (1) FTF while socially distanced and wearing masks (hereafter, “face-to-face” or “FTF”); (2) VC in a laboratory setting that allowed each partner to see one another’s full torso (hereafter, “laboratory VC”); or (3) VC in a remote setting that allowed each partner to see only one another’s face and shoulders (hereafter, “remote VC”). In order to also investigate the effect that these changes have on different kinds of conversations, each pair held three different types of conversation: (a) an affiliative conversation; (b) an argumentative conversation; and (c) a cooperative task-based interaction. We quantified interpersonal coordination by extracting overall amounts of movement from videos of the participants using well-validated computer vision methods (e.g., Paxton & Dale, 2013b; Schmidt et al., 2014) and then calculated the relationship between the two participants’ movement using nonlinear time series analyses (e.g., Paxton & Dale, 2017; Romero et al., 2016; Shockley, 2005).

1.4.1. Hypotheses

We approached the study with five hypotheses (see Table 1). First, participants in the FTF condition should demonstrate the highest amount of overall interpersonal coordination (measured using nonlinear methods explained in detail below), and participants in the remote VC condition should demonstrate the lowest amount of interpersonal coordination (H1). Further, the differences in the amount of interpersonal coordination should be reliably related to the amount of visual information available to participants about their partner (H2). Based on previous research (Paxton & Dale, 2013a, 2017), the argumentative conversation should be characterized by the lowest amount of interpersonal coordination (H3). Given the reduction in psychological closeness and immediacy during VC compared to FTF (e.g., Mallen et al., 2003; Sprecher, 2014), the emotional impact of each conversation type should be dampened in VC interactions (H4). Finally, given the differences in personal investment across interaction contexts (noted for H4), the argumentative and affiliative interactions’ outcomes should show greater differences than the task-based interaction’s outcomes between VC and FTF conditions (H5).

1.5. Potential applied impacts

In addition to shedding new light on the basic processes of human interaction, the current research may have applied impact in the short and long term. Although the COVID-19 pandemic abatement measures have largely faded by 2023, VC will likely continue to play a larger role in our social and work lives than it did before the pandemic (Leonardi,

2021). This work will provide empirically supported recommendations for collaboration, socialization, and even confrontation over VC. Additionally, if future pandemics utilize similar pandemic abatement measures, our findings could provide guidance to improve wellbeing during periods of social distancing.

2. Method

2.1. Participants

One hundred thirty-eight (138) people participated in pairs. Of these, 23 pairs were in the FTF condition, 23 pairs in the Laboratory VC condition, and 23 pairs in the Remote VC condition. Participants were recruited from a small liberal arts college in the U.S. All participants were 18 years of age or older and were fluent in English. Information about exclusion criteria and replacement is presented in Section 2.1.4.

2.1.1. Demographic makeup of participants

Participants were 18 to 22 years old ($M = 19.83$, $SD = 1.27$). In terms of gender, 68.12 % of participants identified as female, 27.54 % identified as male, 3.62 % identified as nonbinary/genderfluid/gender non-conforming, and 0.72 % identified as transmasculine. In terms of race and ethnicity, 59.42 % of participants identified as White, 22.46 % identified as Asian, 5.80 % identified as multiracial, 4.35 % identified as Latinx/Hispanic, 4.35 % identified as Black/African-American, 0.72 % identified as Native Hawaiian or Pacific Islander, 0.72 % identified as Middle Eastern, 0.72 % identified as European, 0 % identified as Native American, and 1.45 % preferred not to answer the question. For these demographic questions, participants were permitted to select multiple options and/or provide their own identities in their own words.

Participants also indicated if they knew their partner before the experiment. Of these participants, 9.42 % stated that they had known their partner before participating in the study, 13.77 % that they were not sure or that they maybe knew the other participant, and 76.81 % that they had no knowledge of the other before the study. Of those who knew each other previously, 7.97 % indicated having met the other once or twice before, 2.90 % indicated that they were acquaintances, 7.25 % indicated that their partner was a classmate, 3.62 % indicated that they were friends before the study, 0.72 % indicated that their partner was their best friend, and 0.72 % indicated that the other was a romantic partner.

2.1.2. A priori power analysis

We determined our sample size by conducting simulation-based power analyses using pwr (Champerley, 2020), lme4 (Bates et al., 2015), and simr (Green & Macleod, 2015) libraries in R (R Core Team, 2020). In keeping with traditional expectations of statistical significance in our field, we assumed an alpha criterion of 0.05; to ensure appropriate power, we assumed power of 80 %. Based on previous research about the impact of interaction context in interpersonal coordination dynamics, we assumed medium effect sizes (e.g., Paxton & Dale, 2013a, 2017; Tschacher et al., 2014).¹

2.1.3. Ethical approval

This project was reviewed and approved by the institutional review board at the college at which it was conducted.

Table 1
List of hypotheses and expected results.

Hypothesis	Expected results
H1 & H2	Level of interpersonal coordination: FTF > laboratory VC > remote VC
H3	Level of interpersonal coordination: Affiliative > coordinative > argumentative
H1-H3	Level of interpersonal coordination: FTF affiliative > > VC argumentative
H4	Emotional impact of conversation: FTF > (laboratory VC remote VC)
H5	Emotional impact of conversation: FTF affiliative > FTF cooperative > VC affiliative > VC cooperative > FTF argumentative > VC argumentative

¹ After our analyses (see Section 4), we conducted post hoc power analyses using simr (Green & Macleod, 2015) in R (R Core Team, 2020), as proposed in our Stage 1 Registered Report (Romero & Paxton, 2021). Because post hoc power analyses are tightly coupled to statistical significance (e.g., O’Keefe, 2007; Yuan & Maxwell, 2005), we include these analyses in our online supplementary materials (see Section 3.6).

2.1.4. Exclusion and replacement

In keeping with our Stage 1 Registered Report (Romero & Paxton, 2021), we removed and replaced all dyads who did not complete the full experimental procedures, who withdrew their consent, whose data were lost due to technical difficulties, or whose data were unusable due to experimenter error. Details about our pre-registered exclusion criteria, along with the number of pairs excluded in each case, can be found in Appendix A.

2.2. Materials and conditions

2.2.1. Pre-study opinion survey

Participants were asked to individually fill out a brief survey on Qualtrics before their study session regarding their position and opinion on a series of current topics (see Table 2 for a complete list of questions; modified from Paxton & Dale, 2013a). Additionally, they indicated how strongly they held their typed opinion on the subject using a 4-point Likert-type scale, with 1 indicating “feel very weakly” and 4 indicating “feel very strongly.”

2.2.2. Affiliative conversation

The affiliative conversation followed procedures similar to previous work (Paxton & Dale, 2013a, 2017). Participants were asked to talk about things that they enjoyed (e.g., hobbies, movies, music), to find something that they both enjoyed, and to discuss why they enjoyed it.

2.2.3. Argumentative conversation

The argumentative conversation followed procedures similar to previous work (Paxton & Dale, 2013a, 2017). Each dyad was asked to discuss one of the topics from the opinion survey for 8–10 min. To elicit a truly argumentative conversation, topics were selected from the opinion questions by identifying topics on which the participants in that dyad wrote differing opinions but for which they each indicated feeling somewhat strongly (i.e., 3) or very strongly (i.e., 4) about their opinion. Again, following previous procedures, if participants stopped talking or came to a resolution on the topic before 8 min had elapsed, the

experimenter stopped the conversation and assigned the second-highest ranking topic as the new prompt for the remainder of the 8- to 10-min time.

2.2.4. Cooperative task-based interaction

The task-based interaction asked participants to work together to identify objects that they would take to a desert island if they had a choice and were going together. They were instructed that all the objects that they picked must neatly fit inside a shoe box. Like both conversations, the participants had 8–10 min to complete the task-based interaction. If participants stopped talking or came to a resolution, the experimenter stopped the conversation and instructed them to find a new set of items (under the same conditions) that did not include any of the previous items.

2.2.5. Interaction surveys

After each interaction, each participant individually rated their agreement with three statements about their partner during the conversation that they just finished. These statements were: “I felt close to my partner;” “I felt that my partner understood what I was saying;” and “I felt that I understood what my partner was saying.” They indicated their level of agreement on a 6-point Likert-type scale, with 1 indicating “very strongly disagree” and 6 indicating “very strongly agree.” These surveys were completed using Qualtrics.

2.2.6. Demographic survey

At the end of the study session, participants indicated their age; their gender identity/identities; their racial/ethnic identity/identities; and whether they knew their partner prior to the study session. If they indicated that they knew their partner prior to the study, they were asked to specify the kind of relationship they had from the following options: *I’ve met them once or twice before*; *Acquaintance*; *Classmate*; *Friend*; *Best friend*; *Romantic partner*; or *Relative*. This survey was also completed through Qualtrics.

2.3. Procedure

Participants individually signed up for one of the three possible study locations (described below) in an automated system. Once they signed up for a date and time to complete the study, they received an email asking them to complete the pre-study opinion survey at least 3 h before their scheduled time. After this, participants completed one of the following procedures, in accordance with their communication medium. For all conditions, conversation order was counterbalanced to avoid order effects, and participants were not told what kinds of conversations they would be having before they received each prompt.

2.3.1. Face-to-face

Pre-interaction procedures.

Participants went to a laboratory space on campus. They were asked to follow CDC and campus-wide preventative measures, such as wearing face coverings and using hand sanitizer. Once at the study location, they were directed to two separate small rooms in order to complete the informed consent process online. After the informed consent process, they were directed to a large room where the conversations took place. During their conversations, participants were seated facing each other 6 ft apart (182 cm; per CDC guidelines). A hanging microphone was placed between them to capture the audio of their conversation. Additionally, a video camera was placed 480 cm away and captured both participants in the same frame.

2.3.1.1. Interaction procedures. Participants were asked to briefly introduce themselves to each other for 2 min without the experimenter in the room. The experimenter then returned to the room and started with the first conversation. Once 8–10 min passed, participants were

Table 2
List of questions included in the pre-study survey from which topics of the argumentative and affiliative conversations will be picked.

Argumentative questions	
1	How do you feel about the death penalty in the U.S.?
2	Do you think that people should be required to wear masks all the time during the COVID-19 pandemic?
3	How do you feel about in-person education during the COVID-19 pandemic?
4	Do you think that misinformation and disinformation are impacting the U.S.?
5	Do you think that the legal drinking age in the U.S. should be lowered from 21 to 18 years old?
6	Do you think marijuana should be legalized nationally in the U.S.?
7	Do you think rich Americans should be taxed more?
8	Do the Kardashians (or other celebrities) deserve their fame?
9	Do you think voting in government elections is important?
10	Should people be able to video record strangers in public?
11	Should people with criminal records be allowed to vote?
12	Should social media sites be moderated?
13	Should you raise children within traditional gender norms using certain toys and allowing certain sports for example?
14	Should celebrities who are popular with children have the responsibility of being positive role models?
15	Should people have the “right to be forgotten” (remove their information from internet search engines and archives)?
Affiliative questions	
16	Name your 5 favorite movies of all time.
17	Name your 5 favorite movies that have premiered in the last 3 years.
18	Name your 5 favorite TV shows that have started in the last 3 years.
19	Name your 5 favorite TV shows of all time.
20	Name your 5 favorite bands or musicians.
21	What is your favorite hobby?

taken back to their original small rooms to complete the interaction survey. After the surveys were completed, they returned to the large room to complete the second conversation. This procedure was repeated for the second and third conversations. Along with the interaction survey from the third conversation, participants were also asked to complete the demographic questions.

2.3.1.2. Post-experiment debrief. After completing the interaction procedures, a member of the research team debriefed the participants on the specifics of the study and answered any questions they had.

2.3.2. Laboratory VC. Pre-interaction procedures

Participants convened at a laboratory space on campus. They were placed in separate rooms with identical setups, each equipped with a desktop computer (Dell All-in-One with built-in speakers), webcam with a built-in microphone (Microsoft LifeCam Studio), and one projector. Zoom (<https://zoom.us>) was used as the VC software and service. They first completed the informed consent process electronically and were given the opportunity to ask any clarifying questions. They were then directed to sit in a chair in the middle of the room in front of a wall on which the video feed of their partner was projected. No self-view was projected, only that of their partner. The projector was placed 295 cm away from the wall, creating a projected image 129.5×200.6 cm on the wall. The webcam was placed by the wall in front of the participant at a distance between 109 and 177 cm, ensuring that their whole torso was visible in the video frame. Once both participants were situated in their respective rooms, the experimenter closed the doors to those rooms. The experimenter then sat in a third room from which they provided instructions through the video call with no video feed. Once everyone was situated, the experimenter told participants to remove their face-mask if they felt comfortable doing so since they were in a building with good air circulation and separated from any other person physically.²

2.3.2.1. Interaction procedures. The experimenter asked participants to briefly introduce themselves for 2 min with the experimenter's audio turned off. The experimenter then un-muted themselves, gave the participants the prompt for their first conversation, re-muted themselves, and started recording the call; this procedure was used for each prompt assignment. Once 8–10 min had passed, the experimenter stopped the recording, un-muted themselves, and directed the participants to move to the desktop to complete the interaction survey; participants were told to return to the chair in the middle of the room when they were done with the survey. This procedure was repeated for each of the three conversations, with the addition of the demographic questions at the end of the third post-conversation questionnaire.

2.3.2.2. Post-experiment debrief. After this had been completed, the participants were directed to put their face mask back on and to walk out of the individual rooms into a shared physical space. A member of the research team then debriefed them on the specifics of the study and answered any questions they had.

2.3.3. Remote VC. Pre-interaction procedures

Once participants signed up to participate in the study, they received an email (reproduced in Appendix B) with specific instructions on how to prepare their computer setup for the study. Specifically, they were asked to: (1) place their computer (or camera) on a stable surface (such as a desk or table, not their bed or legs) to minimize movement of the computer to avoid changing camera placement; (2) sit in front of a wall such that there is no movement behind them, ensuring that the

movement extracted from the video stream is of the participant only; (3) keep their camera stream on for the entirety of the study session; and (4) turn off their self-view. Finally, they were emailed the link to the VC call room that would be taking place; to improve safety, a new room was created for each dyad. As with the lab VC condition, we again used Zoom (<https://zoom.us>) as our VC software. Once the study time started, a member of the research team started the prearranged video call from one of the laboratory computers. The experimenter turned on their video briefly to introduce the study, answer any initial questions, and confirm that the participants were following the setup instructions. Then, they sent the survey link to the participants through the chat function on the VC software and asked them to complete the informed consent portion of the survey. Once completed, participants were directed to come back to the video call to start the experimental session. From this moment on, the experimenter provided instructions through the video call with no video feed.

2.3.3.1. Interaction procedures. The interaction procedures were the same as the Laboratory VC condition, with the exception that the participants held the interactions and completed the surveys on the same computer. For ease of remote participation, participants completed all interaction surveys and demographic information on the same survey platform through which they completed the informed consent.

2.3.3.2. Post-experiment debrief. After this had been completed, the experimenter debriefed them on the specifics of the study over the video call and answered any questions they had. Once all of the questions had been successfully answered, the experimenter ended the video call.

3. Analyses

3.1. Extracting movement

From the video streams, we extracted time-series of participants' movement throughout each conversation using the frame-differencing method (Paxton & Dale, 2013b; Schmidt et al., 2014) implemented with the OpenCV library (Bradski, 2000) in Python (Van Rossum & Drake, 2009). Essentially, this quantified movement through the changes in pixels from frame to frame. Given the laboratory setup in the FTF and Laboratory VC condition and the participant instructions in the Remote VC condition, the only changes in pixel colors from one frame to the next were due to movement by the participants, since the background did not move and no one was allowed to enter participants' experimental rooms during recording. (In keeping with our a priori Exclusion Criteria described in Section 2.1.4, those in the Remote VC condition who did not follow the video setup instructions were excluded from analyses and replaced.) Following previous research practices, the movement time series used for analyses were downsampled to 10 Hz (e.g., Paxton & Dale, 2013a, 2017).

3.2. Quantifying coordination

Once we extracted the movement time series, we performed cross-recurrence quantification analysis (CRQA) for each conversation for each pair using the crqa library (Coco et al., 2020) in R (R Core Team, 2020). CRQA is a powerful nonlinear time series method that is able to quantify not only the level of interconnectedness between two time series but also the patterning and stability of that interconnectedness (e.g., Pellecchia et al., 2005; Shockley et al., 2002). In psychology, CRQA is often used as a measure of interpersonal coordination. As such, it has been used to compare differences in the emergence of coordination according to clinical differences (e.g., Romero et al., 2016; Warlaumont et al., 2010), conversational contexts (e.g., Paxton & Dale, 2017; Shockley et al., 2003), and task demands (e.g., Demir et al., 2019; Ramenzoni et al., 2011). CRQA's flexibility is ideal for comparing

² As described in the "Exploratory Analyses" section (below), this choice would have been analyzed based on participants' choices. However, every participant chose to remove their face covering in this condition.

contexts and constraints that have been relatively understudied—like those targeted here—as it is able to capture linear and nonlinear differences (Carello & Moreno, 2005).

CRQA measures the time-evolving similarity between two time-series by identifying whether behavioral states re-occur over time and then quantifying the dynamic patterns of these recurrences. Major benefits of this analysis include that it does not require the researcher to make any *a priori* assumptions about the structure of the data being analyzed (Romero et al., 2016) and that it is robust to nonstationarity and nonlinearities in the time series (Carello & Moreno, 2005). However, it is critical to note that CRQA metrics do not have inherent meaning (unlike, e.g., fractal measures; Havlin et al., 1995): CRQA metrics are only meaningful as *relative* values, comparing conditions or groups to one another (Coco & Dale, 2014).

From CRQA, we extracted two types of metrics. First, we examined *interaction-level metrics*, which describe the interconnectedness of two time series as a whole. Second, we extracted *moment-to-moment metrics*, which characterize the coupling of movement in a narrower relative window of time across the interaction.

3.2.1. Interaction-level metrics

While numerous metrics can be extracted from CRQA, we targeted three interaction-level metrics. *Recurrence rate* (RR) measures the percentage of recurrent states (relative to total possible recurrent states) between the two time series in question; it can be interpreted as an index of coordination and as the inverse of noise in the system. *Determinism* (DET) measures the percentage of recurrent states that form sequential lines of recurrent points; it quantifies the degree to which co-variation between the two time series is the result of deterministic versus random processes. Finally, the *maximum line length* (MaxLine or maxL) measures the longest sequence of recurrent points; it captures the stability between the coordination of the two time series and can be interpreted as the attractor strength. For more on these and other CRQA measures, see Coco and Dale (2014) and Pellicchia et al. (2005).

3.2.2. Moment-to-moment metrics

The interaction-level metrics of CRQA can uncover subtle patterns across the entire conversation, but we are also interested in understanding the specific patterns around simultaneity. In other words, this closer look allows us to examine how participants' dynamics are coupled from moment to moment, around the timescale of conversation turns. To do this, we will analyze the *diagonal-wise recurrence profile* (DRP), which captures the percentage of recurrent points around simultaneous interaction to identify leading/following, turn-taking, and synchronous interaction patterns. The DRP is similar to a cross-correlation profile, although DRPs are a nonlinear (rather than linear) analysis. As in previous work, we targeted a ± 3 -s window around simultaneity (Paxton & Dale, 2013a, 2017). We then quantified the changes in DRPs with growth curve analyses. For more on this approach, see Main et al. (2016) and Paxton and Dale (2017).

3.2.3. Baselines

To compare the levels of interpersonal coordination observed in our experiment with the levels of interpersonal coordination that might be expected by chance, we created surrogate time series through Fourier phase-randomization analysis (Kantz & Schreiber, 2004; Theiler et al., 1992). Essentially, this method allowed us to respect the properties of the original dataset while disrupting the nonlinearities in the time series. We created 10 surrogate time series for each real time series observed and compared the resulting CRQA metrics to the observed time series' metrics. More information about using this method to create recurrence-based baselines available in Paxton and Dale (2017). Observed levels of coordination were significantly higher than those observed in the baseline; additional information is available in our open materials, data, and code (see Section 3.6).

3.3. Operationalizing outcome

We were interested in how participants' subjective experiences of interactions differ by communication medium and conversation type. As such, interaction outcomes were measured as participants' answers to the post-interaction surveys.³

3.4. Planned analyses

All analyses were conducted in R (R Core Team, 2020).

3.4.1. Interaction-level metrics (H1–H3)

To analyze the interaction-level CRQA metrics, we conducted separate linear mixed-effects models predicting each outcome (RR, DET, and MaxLine) with communication medium (FTF, Laboratory VC, and Remote VC) and conversation type (argumentative, affiliative, and task-based cooperative). Dyad membership was modeled as a random intercept, with the maximal slope permitted to achieve convergence (Barr, 2013; Barr et al., 2013).⁴

3.4.2. Moment-to-moment metrics (H1–H3)

To analyze the DRPs, we conducted a linear mixed-effects model predicting recurrence with orthogonal linear lag (quantifying leading/following), orthogonal quadratic lag (quantifying synchrony or turn-taking), conversation type (argumentative, affiliative, and task-based), and communication medium (FTF, Laboratory VC, and Remote VC). Dyad membership was modeled as a random intercept, with the maximal slope permitted to achieve convergence (Barr, 2013; Barr et al., 2013).

3.4.3. Comparing outcomes (H4–H5)

We conducted linear mixed-effects models predicting each participant's subjective outcome rating with conversation type (argumentative, affiliative, and task-based) and communication medium (FTF, Laboratory VC, and Remote VC). Dyad membership and participant identity were modeled as random intercepts, with the maximal slope permitted to achieve convergence (Barr, 2013; Barr et al., 2013).

3.5. Exploratory analyses

In addition to our planned analyses, we conducted three pre-registered exploratory analyses (E1–E3; Romero & Paxton, 2021). Our first set of exploratory analyses examined the effects of interaction-level and moment-to-moment metrics on subjective outcomes across communication media and conditions (E1). These exploratory analyses were also conducted in R (R Core Team, 2020).

Our second and third pre-registered exploratory analyses were data-dependent (Romero & Paxton, 2021), but sufficient relevant data were not collected. The second would have examined differences between face-visible and face-hidden interactions in the laboratory VC condition if at least one member in at least one-third of dyads in that condition ($n = 7$) had chosen to wear a mask during at least one interaction (E2); however, since every participant (except for one pair) in the laboratory VC condition chose to remove their face mask, E2 was not conducted. The third pre-registered exploratory analysis would have analyzed differences between high-quality and low-quality VC if there had been sufficient variability of reported call quality in VC participants'

³ We recognize that objective outcomes are also important to many kinds of interactions, and we further acknowledge that objective outcomes may differ in important ways from subjective outcomes, even in the contexts we examine here. Future work should examine differences in objective and subjective outcomes.

⁴ Initially, we had planned to use two-way ANOVAs to compare CRQA outcomes, however, our data violated some of the assumptions of the test.

interactions in Laboratory VC and/or Remote VC conditions (E3); however, because none of the dyads in any VC condition reported issues with the video or audio, E3 was not conducted.

3.6. Commitment to open science

We are committed to meeting our ethical responsibilities to our participants and to the scientific community. As such, we undertook the following open science measures for the project.

3.6.1. Open methods

We published all scripts and procedures, both in appendices in the current paper and in a project on the Open Science Framework (OSF: <https://osf.io/47jek/>). The OSF project includes a registered version that reflects the state of the project at the time of acceptance. The project has been shared with a Creative Commons 4.0 Attribution-ShareAlike (CC 4.0 BY-SA) license to promote reuse.

3.6.2. Open data

Due to the inherently identifiable nature of these data, we are not able to share the raw video and audio data. However, we have shared the extracted movement time-series and de-identified participant response data on a restricted-access repository on the Inter-university Consortium for Political and Social Research (ICPSR). We have also shared an analysis-ready dataframe in the OSF project, including only processed and de-identified data. The data have been shared with a CC 4.0 BY-SA license.

3.6.3. Open code

All code used for data collection, data preparation, and data analysis have been published in a public repository on GitHub. The GitHub repository (https://github.com/a-paxton/visual_information_in_coordination) is attached to the OSF project for ease of access and for inclusion in the registered version of the project. The code has been shared with an MIT open-source license to promote reuse.

4. Results

4.1. CRQA parameters

In order to conduct CRQA over our continuous data, we first needed to conduct *phase-space reconstruction* (PSR). Essentially, PSR allows us to use the observed data from one part of a multidimensional interacting system in order to approximate the behavior of the *full* system. From that, we can identify the path of the system as it visits and re-visits states across its multidimensional landscape. Through PSR, we identify two essential parameters for CRQA: *delay* (essentially, a means of using copies of the observed variable to provide unique information about the other variables in the full system) and *embedding dimension* (essentially, a means of identifying how many copies of the observed variable are needed to approximate the full system). A full accounting of the rationale and process of PSR is outside of the scope of the current work, but interested readers may find [Coco and Dale \(2014\)](#) and [Shockley \(2005\)](#) helpful.

For the sake of the current work, it is useful to note that PSR is conducted over each *individual* time series. In our case, we have two participants in each dyad, and each dyad holds three conversations; in total, this means that each dyad contributes a set of six time series to our dataset. Running PSR across each time series could yield very different delays and embedding dimensions. Previous research has taken one of two approaches to resolving this: One approach looks at the individual parameters of each time series to identify “good enough” parameters for the whole dataset (*global parameters*), while the other approach allows each time series to “live” in its own parameters (*tailored parameters*).

Here, we took a balance of the two approaches. After conducting PSR over each time series, we identified two sets of global parameters from

the tailored parameters for each time series. We then identified a single *radius* for each set of parameters (i.e., two sets of global parameters plus one set of tailored parameters) that would yield recurrence rates that are reasonable for biological systems (approximately 1–5 RR); the radius is used to identify what counts as re-visiting a state, since continuous multidimensional systems will likely never return to *precisely* the same value over time. In keeping with previous literatures, we then examined whether the distributions of the CRQA metrics differed across each set of parameters. As presented in our open materials (see [Section 3.6](#)), the results were generally stable across all sets of parameters. Given this stability—and before viewing any results by communication medium or conversation type—we chose the set of parameters that yielded (for our entire dataset) the most reasonable set of RR values for biological systems: a delay of 3, an embedding dimension of 14, and a radius of 0.2. Our other parameters—as well as results from each—are available in our open materials.

4.2. Interaction level metrics

For clarity and completeness, we use tables to present all results of all analyses run. For our readers, we do not include any test statistics in the text; instead, we point our readers to the appropriate tables for additional detail. Again, all models used the affiliative conversation and FTF condition as reference categories for categorical comparisons.

4.2.1. Recurrence rate (coordination or inverse of noise)

After examining the model fit, the planned model predicting RR did not have normally distributed residuals, so we log-transformed RR prior to including it in the model. Compared to affiliative conversations, both argumentative and cooperative conversations showed more coordinated and less noisy movement signals (i.e., significantly higher log RR; see [Table 3](#) and [Fig. 1](#)). No other differences reached statistical significance.

4.2.2. Determinism (stability)

When compared to affiliative conversations, cooperative and argumentative conversations led to more deterministic (or stable) interpersonal coordination (see [Table 4](#) and [Fig. 2](#)). Additionally, interpersonal movement dynamics were significantly more structured in Laboratory VC interactions than in FTF interactions (see [Fig. 2](#)). No other significant differences were present.

4.2.3. Maximum line length (attractor strength)

Participants showed significantly longer bouts of stability in their interpersonal movement when having argumentative conversations than affiliative conversations (see [Table 5](#) and [Fig. 3](#)), suggesting a stronger attractor toward movement synchrony during those conversations. No other differences were present.

Table 3

Linear mixed-effects model predicting Log RR with conversation type (affiliative, argumentative, or cooperative), and condition (face-to-face, VC remote, or VC laboratory). Dyad included as a random intercept. Affiliative conversation and FTF conditions are coded as reference categories. Bolded are significant * at the .05 level and ** at the .001 level.

Predictors	Estimates	CI	p
Intercept	−0.25	−0.82–0.32	0.388
Arg.	0.87	0.34–1.39	0.001
Coop.	0.75	0.23–1.27	0.005
VC remote	−0.01	−0.82–0.80	0.980
VC lab	0.74	−0.07–1.54	0.075
Arg. × VC remote	−0.27	−1.00–0.47	0.476
Coop. × VC remote	0.20	−0.54–0.93	0.601
Arg. × VC lab	−0.25	−0.98–0.49	0.511
Coop. × VC lab	−0.45	−1.19–0.28	0.226
Marginal R ² /conditional R ²	0.086/0.620		

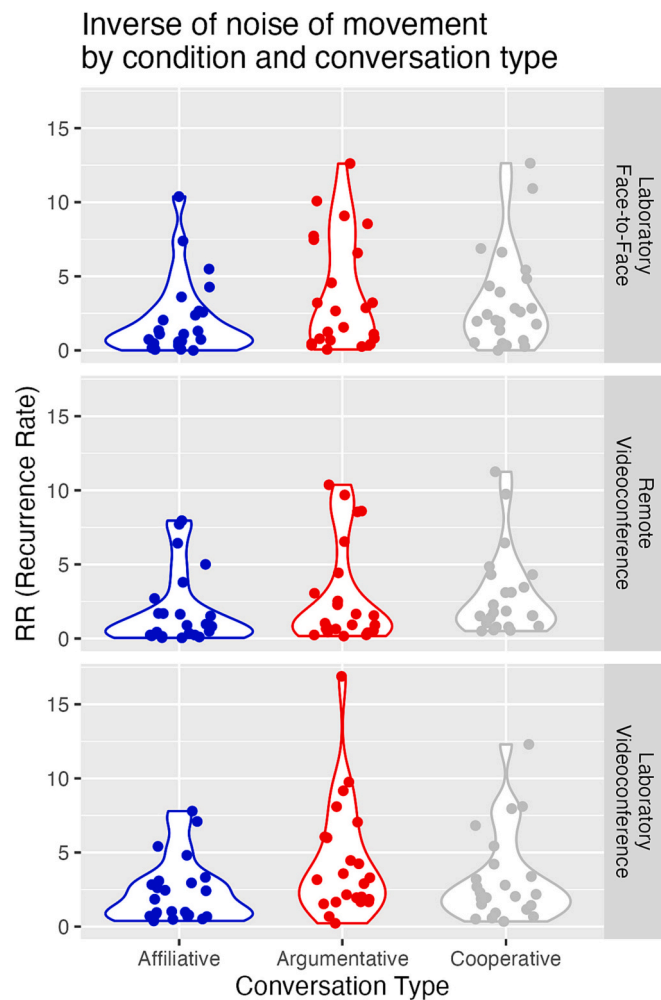


Fig. 1. Violin plots of RR by conversation type (affiliative to the left in blue, argumentative in the middle in red, and cooperative to the right in gray) and condition (laboratory FTF at the top, remote VC in the middle, and laboratory VC at the bottom). Plotted data include *observed* RR; analyses used *log-transformed* RR. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 4

Linear mixed-effects model predicting DET with conversation type (Affiliative, Argumentative, or Cooperative), and condition (Face-to-Face, VC Remote, or VC Laboratory). Dyad included as a random intercept. Affiliative conversation and FTF conditions are coded as reference categories. Bolded are significant * at the .05 level and ** at the .001 level.

Predictors	Estimates	CI	p
Intercept	94.88	92.89–96.87	<0.001**
Arg.	2.84	0.68–5.00	0.010**
Coop.	2.47	0.31–4.63	0.025*
VC remote	2.47	–0.34–5.29	0.084
VC lab	3.31	0.50–6.13	0.021*
Arg. × VC remote	–2.02	–5.07–1.04	0.194
Coop. × VC remote	–1.14	–4.20–1.91	0.461
Arg. × VC lab	–1.98	–5.03–1.07	0.202
Coop. × VC lab	–1.86	–4.92–1.19	0.230
Marginal R ² /conditional R ²	0.057/0.445		

4.3. Moment-to-moment metrics

Because we had no a priori expectations about how leader/follower differences would emerge within specific dyads, we transformed the data so that any participant of a given dyad who tended to “lead” the

movement coordination in a conversation would always be on the same side of the DRP. This allowed us to capture the degree to which *any* leader emerged in the conversation (e.g., Paxton & Dale, 2013a, 2013b), rather than having location- or role-based expectations about whether a *specific* leader emerged. This stands in contrast to studies that *do* have such role- or location-based expectations (e.g., keeping people assigned to a specific role in a role-sensitive task on a specific side of the DRP in Schwab et al., 2022; keeping mothers on one side of a DRP and children on the other side of a DRP in mother-child dyads in Main et al., 2016); in those cases, analyses of DRPs typically preserve those features of the interactions in order to test *specific* hypotheses about who might lead or follow.

Several main effects emerged (see Table 6 and Fig. 4). Cooperative conversations showed higher interpersonal synchrony than affiliative conversations. All conversations tended to be characterized by leader-follower movement dynamics (again, as captured by orthogonal linear lag) and by synchronous movement dynamics (again, as captured by orthogonal quadratic lag).

We also found a number of interaction effects—although not always in the expected direction. Compared to FTF interactions, both remote VC and lab VC were marked by significantly more synchrony. Compared to affiliative FTF conversations, we found significantly higher synchrony in argumentative remote VC conversations and cooperative lab VC conversations. Cooperative remote VC conversations also had significantly more pronounced leader/follower movement dynamics. No other effects reached statistical significance.

4.4. Comparing outcomes

Linear mixed effects models revealed no significant differences in subjective ratings of closeness, subjective ratings of understanding their partner, or subjective ratings of feeling understood by their partner based on condition or type of conversation (see Table 7).

4.5. Exploratory analyses

We conducted an iterative model-building process in order to explore the impacts of movement coordination *during* the conversation and experimental interaction constraints on the resulting subjective outcomes *after* the conversation. To do so, we created separate sets of models for each outcome (i.e., feelings of closeness, of understanding their partner, and of feeling understood by their partner). As a baseline, we used the linear mixed-effects models predicting each outcome with conversation type, condition, and their interaction (with random intercepts of dyad and participant). From there, we built models that added qualifications of the interaction dynamics: log RR, DET, MaxLine, and the maximum RR observed in the DRP.⁵ Because these metrics have dramatically different ranges, we centered and scaled the interaction dynamics prior to entering them into the models. We then compared the models to the baseline model to determine whether they better described the data.

None of the models were significantly better fits better than the baseline model. In other words, the constraints of the conversation alone were the best predictor of participants' subjective experiences of the conversation.

4.5.1. Exploring conversation order as a constraint

Our results thus far underscored the importance of conversational constraints on the emergent dynamics. However, we were surprised by the pattern of results, given the previous findings that guided our

⁵ Essentially, this captures the maximum amount of moment-to-moment coordination within the ± 3 -second window of our DRPs. While this therefore does not embrace the full variability of the DRP, it provides a more tractable slice of the DRP for this modeling.

Structure of movement by condition and conversation type

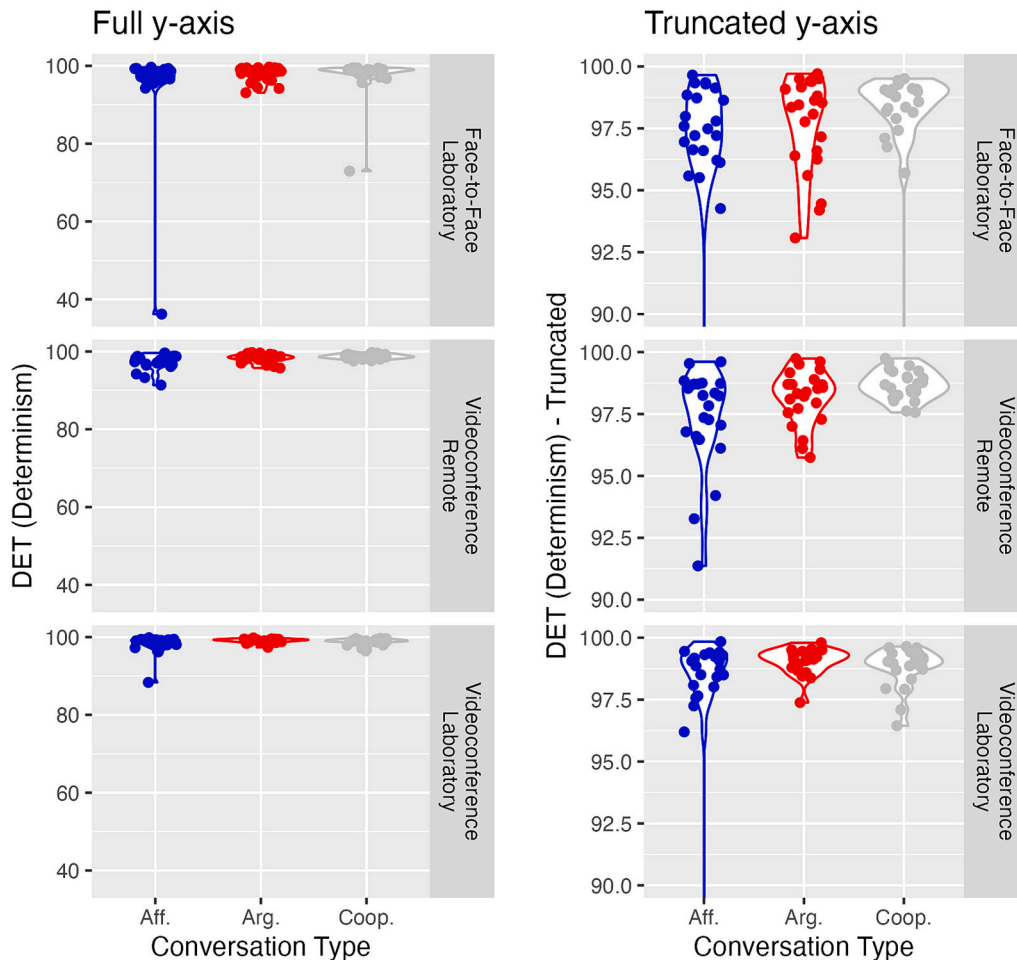


Fig. 2. Violin plots of DET by conversation type (affiliative to the left in blue, argumentative in the middle in red, and cooperative to the right in gray) and condition (laboratory FTF in the top panel, remote VC in the middle panel, and laboratory VC at the bottom panel). Includes full y-axis (left) and truncated y-axis (right) to provide complete and contextualized data visualizations. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 5

Linear mixed-effects model predicting MaxLine with conversation type (Affiliative, Argumentative, or Cooperative), and condition (Face-to-Face, VC Remote, or VC Laboratory). Dyad included as a random intercept. Affiliative conversation and FTF conditions are coded as reference categories. Bolded are significant * at the .05 level and ** at the .001 level.

Predictors	Estimates	CI	p
Intercept	115.57	87.22–143.91	<.001**
Arg.	37.35	9.03–65.66	0.010**
Coop.	11.22	–17.10–39.53	0.436
VC remote	–10.61	–50.69–29.48	0.602
VC lab	23.09	–17.00–63.17	0.257
Arg. × VC remote	–7.00	–47.04–33.04	0.731
Coop. × VC remote	19.43	–20.61–59.48	0.340
Arg. × VC lab	8.52	–31.52–48.56	0.675
Coop. × VC lab	–12.65	–52.69–27.39	0.534
Marginal R ² /conditional R ²	0.088/0.545		

expectations. We identified a number of possible additional factors that may have influenced our results that we would have been unable to address in the current dataset (described in Section 5), but we looked again at previous work to identify whether we might be able to identify other factors *within* our dataset that could be shaping interpersonal dynamics.

In our search, we were reminded of the non-significant trend toward an effect of conversation *order* on interpersonal synchrony in Paxton and Dale (2013a): Synchrony during argumentative conversations was higher when it was the first conversation that the pair had than when it

was the *second* conversation they had (preceded by an affiliative conversation). One suggested explanation for that trend was the importance of the history of the dyad—that is, argument was more disruptive to the interpersonal system when it violated the friendly basis of the participants' relationship to one another (given that nearly all were strangers). Coupled with other work showing that increased synchrony is associated with increased liking (e.g., Hove & Risen, 2009; Hu et al., 2022; Paxton & Dale, 2013b), we wanted to test whether conversation order might play a role. In other words, can we predict participants' feelings of closeness through participants' emergent movement dynamics *and* the constraints from which those dynamics emerged?

To do so, we created an exploratory model predicting ratings of closeness with log RR, DET, conversation type (i.e., affiliative, argumentative, or cooperative), conversation number (i.e., first, second, or third), and condition (i.e., FTF, VC Remote, and VC Lab). We again centered and standardized log RR and DET prior to entering them into the model and included both participant and dyad as random intercepts. This new model was significantly better than our baseline model of closeness.

The results of our exploratory model support the importance of considering conversation order or number as one of the constraints shaping interpersonal synchrony (see Table 8 and Figs. 5–6). Two of the highest-order interaction terms reached statistical significance, *both* with argumentative conversations in the VC Lab condition (again, compared to FTF affiliative conversations). Although levels of liking tended to increase over time for more stable shared movement patterns (higher DET) for FTF affiliative conversations, levels of liking in

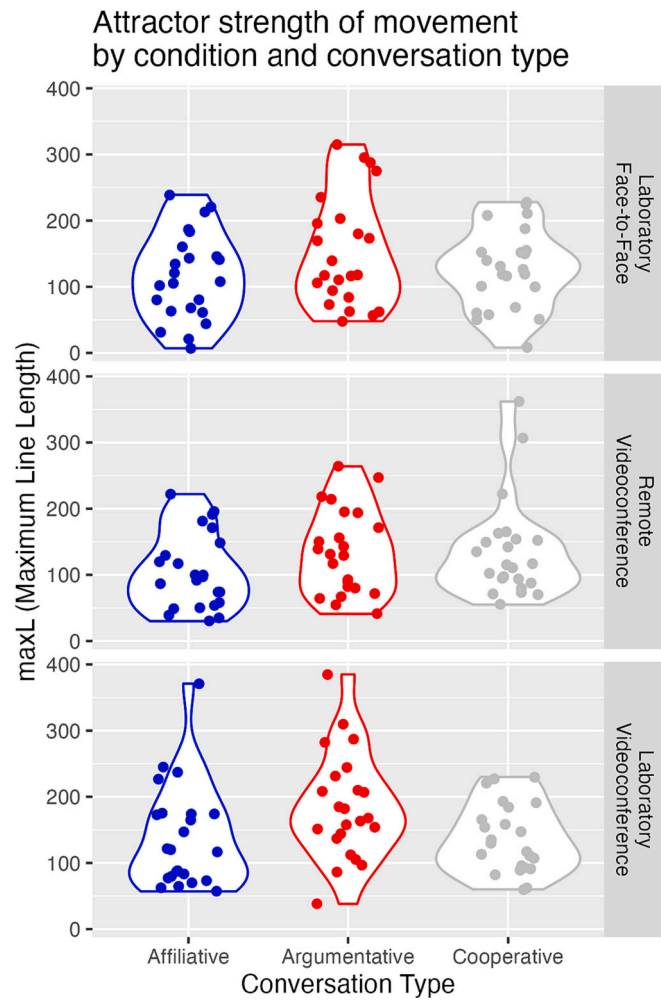


Fig. 3. Violin plots of MaxLine by conversation type (affiliative to the left in blue, argumentative in the middle in red, and cooperative to the right in gray) and condition (laboratory FTF at the top, remote VC in the middle, and laboratory VC at the bottom). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

argumentative conversations in the VC Lab were similar across DET and conversation number. The patterns between the two conditions and conversation types diverged even more when considering the interactions with log RR. Taken together, the patterns of results were consistent with the hysteresis (or history) effects suggested by Paxton and Dale (2013a): Differences in the dyad's interaction history and levels of movement synchrony are related to differences in feelings of closeness, although they are—in this case—also tied to differences in communication medium.

5. Discussion

Within the context of a global pandemic that radically altered our social landscape, we here investigated how the medium and goal of our conversations can change our patterns of physical and social connection. The current study examined the degree to which interpersonal movement coordination (i.e., the similarity of the movement between people) is impacted by the amount of visual information available about conversation partners' movement via different communication media and the type of interaction conversation partners are having. We compared three types of conversations—affiliative, argumentative, and cooperative—held in one of three communication media: face-to-face conversations held while participants were co-located in a laboratory, wore

Table 6

Linear mixed-effects model of diagonal recurrence profiles (DRPs) predicting RR with orthogonal linear lag (lead/follow), orthogonal quadratic lag (synchronization), conversation type (Affiliative, Argumentative, or Cooperative), and condition (Face-to-Face, VC Remote, or VC Laboratory). Dyad and participant included as random intercepts. Affiliative conversation and FTF condition are coded as reference categories. Bolded are significant * at the .05 level and ** at the .001 level.

Predictors	Estimates	CI1	p
Intercept	0.02	0.01–0.03	0.001**
Linear lag	−0.02	−0.03 to −0.01	<0.001**
Quadratic lag	−0.01	−0.01 to −0.00	<0.001**
Arg.	0.01	−0.00–0.02	0.130
Coop.	0.02	0.00–0.04	0.033*
VC remote	0.01	−0.01–0.03	0.349
VC lab	0.01	−0.01–0.03	0.271
L. lag × Q. lag	0.02	−0.01–0.06	0.227
Linear lag × Arg.	−0.00	−0.01–0.01	0.529
Linear lag × Coop.	−0.00	−0.01–0.01	0.597
Quadratic lag × Arg.	−0.01	−0.01–0.00	0.086
Quadratic lag × Coop.	−0.00	−0.01–0.00	0.473
Linear lag × VC remote	−0.00	−0.01–0.00	0.335
Linear lag × VC lab	−0.01	−0.01–0.00	0.260
Quadratic lag × VC remote	−0.02	−0.03 to −0.01	<0.001**
Quadratic lag × VC lab	−0.02	−0.03 to −0.02	<0.001**
Arg. × VC remote	−0.01	−0.03–0.01	0.519
Coop. × VC remote	−0.00	−0.03–0.02	0.875
Arg. × VC lab	0.00	−0.02–0.02	0.792
Coop. × VC lab	−0.01	−0.03–0.02	0.636
L. lag × Q. lag × Arg.	0.02	−0.04–0.07	0.494
L. lag × Q. lag × Coop.	−0.01	−0.07–0.04	0.686
L. lag × Q. lag × VC remote	0.03	−0.02–0.09	0.264
L. lag × Q. lag × VC Lab	−0.01	−0.06–0.05	0.798
L. lag × Arg. × VC remote	0.00	−0.01–0.02	0.515
L. lag × Coop. × VC remote	−0.02	−0.04 to −0.01	<0.001**
L. lag × Arg. × VC lab	0.00	−0.01–0.02	0.468
L. lag Coop. × VC lab	−0.00	−0.02–0.01	0.691
Q. lag × Arg. × VC remote	0.02	0.01–0.02	<0.001**
Q. lag × Coop. × VC remote	0.00	−0.01–0.01	0.623
Q. lag × Arg. × VC lab	0.00	−0.01–0.01	0.524
Q. lag × Coop. × VC lab	−0.01	−0.02 to −0.00	0.008**
L. lag × Q. Lag × Arg. × VC remote	−0.03	−0.11–0.04	0.383
L. lag × Q. Lag × Coop. × VC remote	0.06	−0.02–0.14	0.139
L. lag × Q. Lag × Arg. × VC lab	−0.00	−0.08–0.08	0.961
L. lag × Q. Lag × Coop. × VC lab	0.06	−0.02–0.14	0.124
Marginal R ² /conditional R ²	0.048/0.934		

face masks, sat six feet apart, and had full view of their conversation partner's body movements (FTF); videoconference conversations held while participants were separately in locations of their own choosing, did not wear face masks, and had a view only of their conversation partner's head and shoulders (Remote VC); and videoconference conversations held while participants were separately in laboratory locations, did not wear face masks, and had full view of their conversation partner's body movements (Laboratory VC).

Although our hypotheses had been rooted in previous work, our results generally only partially supported or failed to support our hypotheses. Taken together, these findings open new opportunities for investigation and urge additional nuance in our understanding of how context might impact interpersonal coordination.

5.1. Visual information and “naturalism” across interactions (H1–H2)

We found that communication medium affected coordination, but our results failed to support our hypotheses that face-to-face conversations and that more visual information about a conversation partner would increase interpersonal coordination (H1–H2). We had expected the highest coordination in FTF conversations and the lowest in Remote VC. Interestingly, our interaction-level analyses with CRQA found no differences in the overall amount of interpersonal coordination (or inverse of noise; RR) or the attractor strength (MaxLine) based on the amount of visual information available to participants. Contrary to our

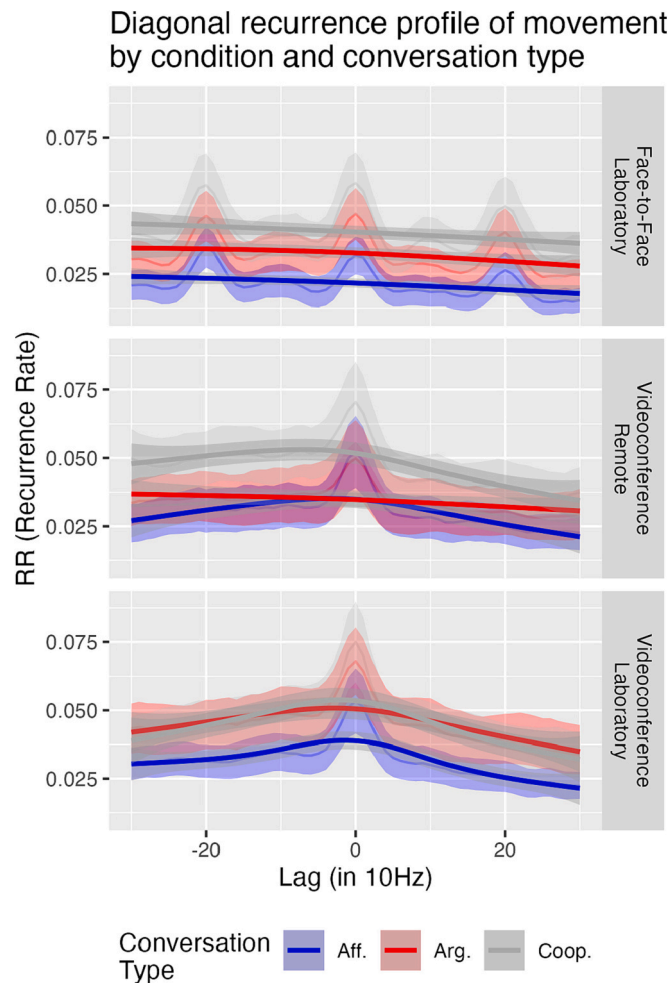


Fig. 4. Diagonal recurrence profile (in RR) of movement by conversation type (affiliative in blue, argumentative in red, and cooperative in gray) and condition (laboratory FTF at the top, VC laboratory in the middle, and VC remote at the bottom) across ± 3 -second lags. Fits for first- and second-order polynomials by conversation type are shown in darker, thicker lines with bands of standard error (SE). Thinner, lighter lines depict raw mean at each time lag by conversation type with lighter bands for raw SE. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

hypotheses, we further found that there was increased stability in interaction-level interpersonal synchrony in the Laboratory VC condition compared to FTF interactions. Additionally, our analyses of DRPs found increased moment-to-moment interpersonal movement

synchrony between participants in both VC conditions when compared to the FTF condition.

These findings were unexpected given previous research. While no work had yet compared the amount of synchrony in FTF and VC at the time of the publication of our Stage 1 Registered Report (Romero & Paxton, 2021), earlier studies had found that greater synchrony emerges when partners have visual access to one another (Richardson et al., 2007; Shockley et al., 2003), that the fluency of interaction changes significantly during technology-mediated conversation (Fussell et al., 2000; Kraut et al., 1996; Schmitt et al., 2018), and that technology-mediated conversation can dramatically alter movement patterns (Socza-wa-Stronczyk & Bocian, 2020; Stoffregen et al., 2004; Tolston et al., 2014; Varlet et al., 2013). Furthermore, since we began our work, Zubek et al. (2022) published findings consistent with our initial hypothesis: They found that dyads' head movements were more stably coordinated during (maskless) face-to-face conversations than during VC conversations.

We have identified several potential reasons why our results may have differed from previous literature and our expectations. First, our FTF condition required participants to wear face masks for COVID abatement procedures. We acknowledged initially that this would, indeed, limit the amount of visual information available to conversation partners, although we may have underestimated the importance of facial information in creating our hypotheses (cf. Section 1.2). Previous work on the degree of visual information on interpersonal synchrony did not distinguish between the importance of any specific *kind* of visual information but solely focused on the *amount* (Richardson et al., 2007; Shockley et al., 2003). However, a host of research in emotion perception, gaze, and conversation underscores the importance of visual information present in facial expressions: Gaze frequently lands on the mouth and nose during interaction (especially in ambiguous settings; Hessels, 2020), attention patterns change from face-to-face and video interactions (Freeth et al., 2013), and people may be differently or even more sensitive to facial expressions than body movements (de Gelder & Solanas, 2021; Kret et al., 2011; Minnebusch & Daum, 2009; Slaughter et al., 2004). By appropriately re-situating the VC Laboratory condition as the condition with the *most* visual information about conversation partners, we may see our results as consistent with the view that visual information is critical to interpersonal synchrony. To test this hypothesis more directly, future work can investigate occlusions to face and body movement more systematically (assuming that pandemic abatement procedures or other health safety measures do not impose restrictions).

Second, pandemic-related changes in the social context may have impacted communication dynamics. Unlike previous studies comparing face-to-face and videoconference conversations, our study brought together participants who faced significant health fears about other people, who found themselves grappling with social isolation, and who were becoming increasingly fluent with using videoconference technologies (e.g., Hawes et al., 2022; Ho & Moscovitch, 2022; Zheng et al.,

Table 7

Linear mixed-effects models predicting post-conversation ratings of closeness (left panel), understood partner (middle panel), and felt understood by partner (right panel) with conversation type (Affiliative, Argumentative, or Cooperative) and condition (Face-to-Face, VC Remote, or VC Laboratory). Dyad and participant included as random intercepts. Affiliative conversation and FTF condition are coded as reference categories. Bolded are significant * at the .05 level and ** at the .001 level.

Predictors	Closeness			Understood partner			Felt understood		
	Estimates	CI	p	Estimates	CI	p	Estimates	CI	p
Intercept	4.48	4.16–4.79	<0.001	5.17	4.94–5.41	<0.001	5.20	4.98–5.41	<0.001**
Arg.	–0.30	–0.61–0.01	0.054	–0.13	–0.35–0.09	0.244	–0.09	–0.29–0.12	0.410
Coop.	–0.24	–0.56–0.08	0.146	0	–0.23–0.22	0.965	–0.02	–0.23–0.19	0.837
VC remote	0.14	–0.30–0.58	0.527	0.19	–0.14–0.52	0.262	0.24	–0.07–0.54	0.126
VC lab	0.14	–0.30–0.58	0.536	0.11	–0.22–0.45	0.503	0.08	–0.22–0.39	0.604
Arg. × VC remote	0.22	–0.21–0.66	0.312	0.11	–0.20–0.42	0.492	0.07	–0.23–0.36	0.662
Coop. × VC remote	0.02	–0.44–0.47	0.944	0.11	–0.20–0.43	0.475	0.11	–0.18–0.40	0.467
Arg. × VC lab	–0.06	–0.49–0.38	0.795	–0.17	–0.48–0.14	0.272	–0.11	–0.40–0.18	0.467
Coop. × VC lab	0.27	–0.18–0.73	0.242	–0.1	–0.42–0.21	0.514	–0.02	–0.32–0.27	0.884
Marginal R ² /conditional R ²	0.034/0.594			0.039/0.489			0.039/0.520		

Table 8

Linear mixed-effects model predicting feelings of closeness with scaled-and-centered interaction-derived metrics (log RR and DET) and experimental constraints (condition, conversation type, and conversation number), including random intercepts of dyad and participant. Determined to be the best-fit exploratory model through model comparison. Additional models are available in the online supplementary materials. Bolded are significant * at the .05 level and ** at the .001 level.

Predictors	Estimates	CI	p
(Intercept)	3.94	3.20–4.67	<0.001**
Log RR	0.56	−0.39–1.51	0.244
DET	−1.33	−3.71–1.06	0.275
Arg.	−0.03	−0.97–0.91	0.953
Coop.	−0.44	−1.66–0.78	0.476
Conv. Num.	0.26	−0.04–0.57	0.094
VC remote	0.17	−0.93–1.27	0.759
VC lab	−1.01	−2.00 to −0.02	0.046*
Log RR × Arg.	−1.30	−2.81–0.22	0.093
Log RR × Coop.	0.71	−1.06–2.47	0.434
DET × Arg.	2.57	−1.40–6.54	0.204
DET × Coop.	−1.78	−8.37–4.80	0.595
Log RR × Conv. Num.	−0.37	−0.85–0.11	0.130
DET × Conv. Num.	0.70	−0.49–1.89	0.249
Arg. × Conv. Num.	−0.10	−0.54–0.34	0.652
Coop. × Conv. Num.	0.11	−0.39–0.61	0.662
Log RR × VC remote	−0.66	−2.29–0.96	0.424
Log RR × VC lab	−1.71	−2.99 to −0.44	0.009
DET × VC remote	0.91	−4.00–5.83	0.715
DET × VC lab	2.83	−0.14–5.79	0.062
Arg. × VC remote	−0.44	−1.87–0.98	0.539
Coop. × VC remote	−0.38	−2.51–1.74	0.724
Arg. × VC lab	3.64	0.90–6.37	0.009
Coop. × VC lab	1.00	−0.78–2.78	0.270
Conv. Num. × VC remote	−0.02	−0.48–0.43	0.918
Conv. Num. × VC Lab	0.70	0.19–1.20	0.007
Log RR × Arg. × Conv. Num.	0.63	−0.23–1.49	0.150
Log RR × Coop. × Conv. Num.	−0.11	−0.89–0.67	0.785
DET × Arg. × Conv. Num.	−1.19	−3.43–1.05	0.298
DET × Coop. × Conv. Num.	0.48	−1.93–2.90	0.695
Log RR × Arg. × VC remote	0.11	−2.14–2.36	0.922
Log RR × Coop. × VC remote	−0.73	−3.63–2.18	0.623
Log RR × Arg. × VC lab	3.45	0.95–5.94	0.007
Log RR × Coop. × VC lab	1.48	−0.80–3.76	0.202
DET × Arg. × VC remote	−0.65	−7.75–6.45	0.857
DET × Coop. × VC remote	4.58	−8.06–17.21	0.477
DET × Arg. × VC lab	−15.83	−29.01 to −2.65	0.019
DET × Coop. × VC lab	0.84	−7.45–9.13	0.842
Log RR × Conv. Num. × VC remote	0.29	−0.48–1.06	0.460
Log RR × Conv. Num. × VC lab	1.09	0.41–1.77	0.002
DET × Conv. Num. × VC remote	−0.30	−2.65–2.05	0.801
DET × Conv. Num. × VC lab	−2.12	−4.14 to −0.10	0.040
Arg. × Conv. Num. × VC remote	0.37	−0.28–1.03	0.262
Coop. × Conv. Num. × VC remote	0.00	−0.91–0.92	0.992
Arg. × Conv. Num. × VC lab	−2.06	−3.32 to −0.80	0.001
Coop. × Conv. Num. × VC lab	−0.57	−1.36–0.23	0.163
Log RR × Arg. × Conv. Num. × VC remote	0.12	−1.07–1.31	0.841
Log RR × Coop. × Conv. Num. × VC remote	−0.12	−1.46–1.21	0.855
Log RR × Arg. × Conv. Num. × VC lab	−2.10	−3.44 to −0.77	0.002
Log RR × Coop. × Conv. Num. × VC lab	−0.95	−2.03–0.13	0.085
DET × Arg. × Conv. Num. × VC remote	−0.76	−4.52–3.01	0.692
DET × Coop. × Conv. Num. × VC remote	−0.37	−5.93–5.19	0.896
DET × Arg. × Conv. Num. × VC lab	9.37	3.02–15.73	0.004
DET × Coop. × Conv. Num. × VC lab	0.66	−2.90–4.22	0.716
Marginal R ² / conditional R ²	0.207 / 0.645		

2020). The upheaval in our participants' lives likely substantially impacted how they behaved within our study (e.g., Gentili & Cristea, 2020). For example, research has long found differences in patterns of interpersonal synchrony according to clinical diagnoses like anxiety and depression (e.g., Asher et al., 2020; Condon & Ogston, 1966; Paulick et al., 2018). At the same time, much of the previous related work on the potential disruption in technology-mediated conversations (e.g., Schegloff, 1982; Schmitt et al., 2018) may have been less relevant, given the dramatic changes in familiarity with VC and advances in VC technology. In keeping with this, we found that very few participants in our study—across all conditions—reported having significant VC quality issues, preventing us from even investigating the degree to which VC signal quality might impact the observed dynamics (as included in our Stage 1 Exploratory Analysis plan). To directly test the idea that fluency might be impacting interpersonal synchrony, future work should directly manipulate call quality and perhaps attempt to recruit participants with less familiarity with VC. It may be, for example, that those with more familiarity with VC display different behavioral and coordination dynamics compared with FTF, as has been seen when comparing VR and real-world interactions (Soczawa-Stronczyk & Bocian, 2020; Tolston et al., 2014; Varlet et al., 2013).

Third, there may have been meaningful differences in what might be called the “comfortability” of the communication context (broadly construed) across the different conditions. Both of the laboratory conditions—Lab VC and FTF—may have been uncomfortable in different ways, with the relatively unusual VC setup and the awkwardness of conversations with masks (respectively), not to mention the potential non-naturalism introduced simply by being in a laboratory. By contrast, the Remote VC setting allowed participants to use a more typical VC setup and in a location of their choosing (as long as it conformed to some basic requirements; e.g., static background). Given how sensitive interpersonal synchrony is to even subtle social factors (e.g., Miles et al., 2011), the increased ease in the Remote VC condition may have counteracted the reduction in available visual information about their partner, not to mention the familiarity all of our participants had with that specific type of set-up at the time of data collection (April 2021 to January 2023). Our decision to allow Remote VC participants to participate from a location of their choosing was largely driven by our desire (at the creation of the study) not to require all potential participants to come into the lab at a time in which they might feel unsafe being co-present with others. This, too, may have led to a potential confound of condition: People who tended to sign up to hold a remote VC conversation may have been fundamentally different from those who were willing to come in person. Future work should control for these factors, for example, by having all three communication media in laboratory settings or in a setting of the participants' choosing.

One final (and particularly interesting) point of comparison to previous work centers specifically on Zubek et al. (2022), published after Stage 1 of our Registered Report (Romero & Paxton, 2021). Their study was focused on answering some of the same questions as ours—that is, investigating differences in face-to-face and videoconference conversations to identify whether access to visual information may drive differences in interpersonal synchrony. Their hypotheses were aligned with ours, but unlike ours, their findings supported those hypotheses. In attempting to understand these two sets of findings, it is useful to point out several important differences in our experimental designs. First, all of their participants knew one another prior to the experiment, all being part of the same academic program, while our participants were mostly strangers. Second, communication medium was a *within-subjects* condition, ensuring that there were no self-selection biases or idiosyncratic responses to condition. Third, movement coordination was assessed with pose estimation software and only considered head movement (and not, e.g., posture or facial expressions), unlike our analysis of all movement through frame differencing. Fourth, the conversation topics were chosen freely by the participants, not assigned by the experimenter. Finally, face coverings were not worn during the face-to-face

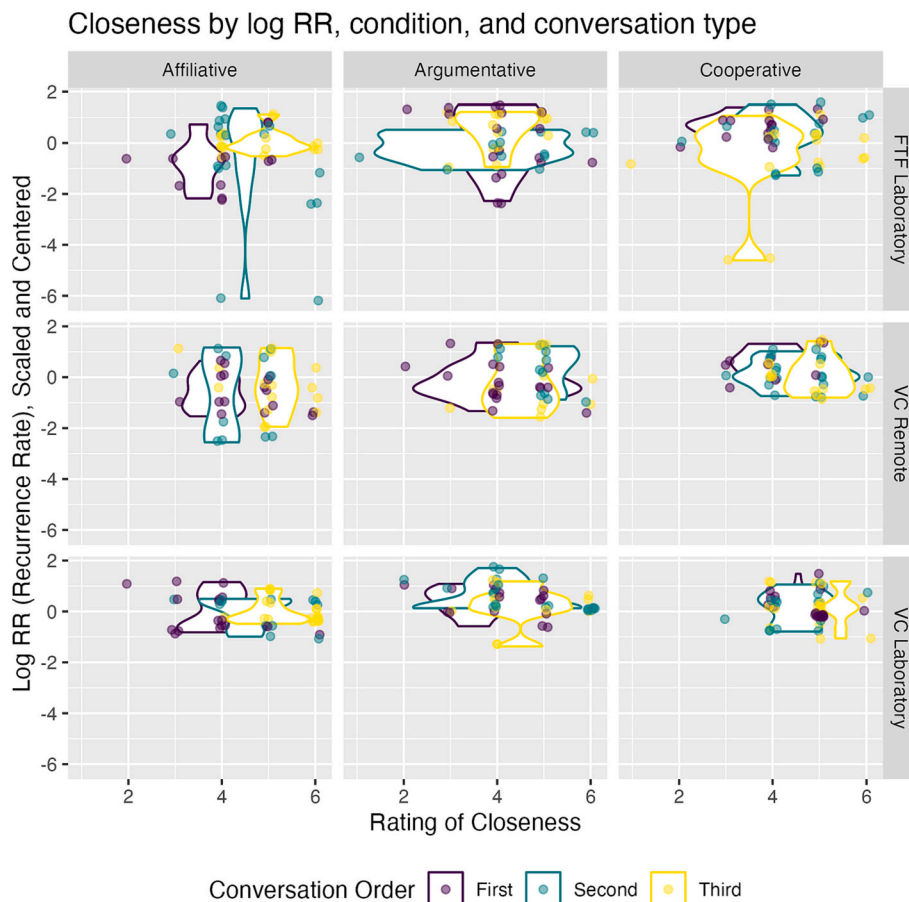


Fig. 5. Violin plots of closeness by log RR (centered and scaled), conversation type (affiliative in left panels, argumentative in middle panels, and cooperative on right panels), condition (laboratory FTF at the top, VC laboratory in the middle, and VC remote at the bottom), and conversation number (first in purple, second in green, and third in yellow). Individual points overlaid in corresponding colors. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

interactions in their study but were required in ours. It is not clear, of course, why *exactly* our findings differ from Zubeck and colleagues', given how many points of difference exist across our two studies. Future work will need to disentangle which factors may have played a role in these differences. However, we take both works to support the general idea that constraining visual information during interaction has interesting and complex consequences in terms of nonverbal communication.

5.2. Conversation type and “connectedness” (H3)

Based on previous research, we predicted that interpersonal movement coordination would be lowest in argumentative conversations (Paxton & Dale, 2013a, 2017) and highest in cooperative conversations (Allsop et al., 2016; Miles et al., 2017). We found only partial support for this hypothesis (H3). Cooperative conversations were marked by sharper synchrony than affiliative conversations (as DRPs). Additionally, moment-to-moment metrics showed argumentative conversations held over Remote VC and cooperative conversations held over Laboratory VC to have higher synchrony than affiliative conversations held FTF. Compared with affiliative conversations, argumentative conversations were overall less noisy or more coordinated than affiliative conversations (as log RR) and had a stronger attractor (as MaxLine). Both cooperative and argumentative conversations were marked by more stability or deterministic movement synchrony than affiliative ones (as DET). In short, we found support for our hypothesis about cooperative conversations but not about argumentative ones. Importantly, we found especially interesting and unexpected effects when considering *both* communication medium *and* type of conversation in terms of synchrony (per DRPs).

Previous work on conflict about closely held opinions—like those studied here—found that interpersonal synchrony decreased compared

to friendly conversations (Paxton & Dale, 2013a, 2017). Given the similarity in our paradigms, we expected similar results in the current work, although we anticipated that the communication medium would interact with them. In considering how and why our results differed from those of previous similar work, we think that an important likely factor is—again—the wider societal context.

During the data collection period for this project (April 2021–January 2023), society shifted rapidly (Centers for Disease Control and Prevention, 2022). After a year of lockdowns and public abatement measures, people in the U.S. saw the initial COVID-19 vaccine become widely available, had a summer of cautious optimism about quashing COVID-19 completely, experienced the resurgence of uncertainty with the emergence of the Delta and Omicron variants, and coped with the slow drawdown of anxiety and uncertainty as COVID became less pandemic and more endemic. The students taking part in our study had lived through a time in which their social lives and schooling had been completely halted or significantly changed. Many felt isolated and searched for deeper social connection, and younger people—like the participants in our study—were some of the most deeply affected (e.g., Hawes et al., 2022; Ho & Moscovitch, 2022). Although we (the authors) went into the study acknowledging that this “new reality” of COVID-19 would likely impact interaction dynamics, it was impossible for us to imagine how much the situation would shift over the following year and a half—and how long its effects would be felt.

Given this, it may be that the broader context of social dynamics fundamentally changed the pressures of the argumentative conversations in comparison to previous work (Paxton & Dale, 2013a, 2017). For example, Lumsden et al. (2012) pointed out that participants with prosocial intentions tend to also show higher interpersonal coordination in certain contexts, and Miles et al. (2011) suggested that people will be more likely to synchronize (and with increased stability) with members

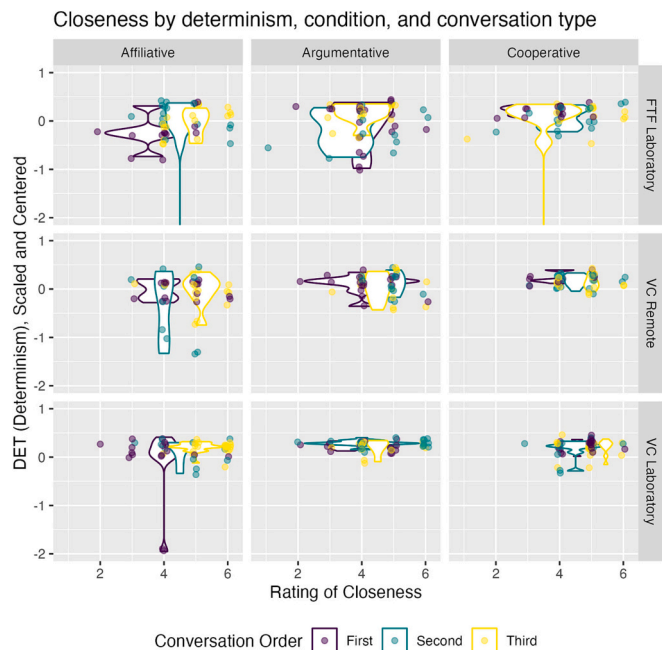


Fig. 6. Violin plots of closeness by DET (centered and scaled), conversation type (affiliative in left panels, argumentative in middle panels, and cooperative on right panels), condition (laboratory FTF at the top, VC laboratory in the middle, and VC remote at the bottom), and conversation number (first in purple, second in green, and third in yellow). Individual points overlaid in corresponding colors. (Note that y-axes are restricted to improve detail.) (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

of other social groups, potentially to bridge differences with people whom they consider to be different from themselves. Whether due to pandemic-related increases in prosocial attitudes (e.g., Bilancini et al., 2022; Syropoulos & Markowitz, 2021), in the drive for social connection (e.g., Ang et al., 2022; Bond, 2021; Dingle et al., 2022), or in the desire for authenticity (e.g., Munn & Weijers, 2021; Wiederhold, 2021), our participants may have experienced even argumentative conversations as fundamentally different than other participants—perhaps as a bridge to other people or as a way of exploring their authentic selves.

Another factor may be the focus on authenticity and authentic connections often emphasized by “Generation Z” in the US (e.g., Ang et al., 2022; Cavanaugh, 2019; Konstantinou & Jones, 2022; Santer, Manago, & Bleisch, 2022), which includes those in our sample. This view of authenticity emphasizes a desire for people to express their true selves, voicing their opinions and connecting with others even about difficult topics. Members of “Gen Z” often define themselves in contrast to previous generations, who—in their view—tend to focus more on preserving their social image (e.g., Boffone, 2022). Arguments about deeply held personal opinions may be seen as opportunities to engage with others authentically—perhaps even more than the affiliative conversation. Importantly, “Gen Z” members have also had the most exposure to video-mediated communication through their preferred social media platforms (Mitchell, 2020), which might also help explain why some VC-mediated conversations showed most synchrony given their ease with and preference of this type of communication. Future research should investigate how differences in social orientation and values (whether or not they stem from generational attitudes) might shape interpersonal dynamics across different conversation types.

One final point to consider may be differences in published work on findings of synchrony within the context of different kinds of conflict. Tschacher et al. (2014) found that interpersonal synchrony increased during a form of gamified competitive debate that used assigned positions, relative to both positive-affect and neutral-affect interactions.

Previous findings suggest that the decrease in interpersonal synchrony during conflict about personally held opinions is not solely attributable to changes in negative emotion during conflict (Paxton & Dale, 2013a). Some of the motivations that we discussed earlier in this section—like pro-sociality, connectedness, or authenticity—may have fundamentally changed the conflict setting in the same way that gamification does. Future work should systematically manipulate different kinds of conflict in order to test how different aspects of the conflict context can reorganize the interpersonal system.

5.3. Conversation outcomes, history, and interacting constraints (H4-H5)

Interestingly, participants' feelings about the conversations they had were not affected by communication medium, type of conversation, or the interaction of those contextual factors (H4–5). In keeping with our planned exploratory analyses, we also tested whether including synchrony metrics would help better explain patterns in subjective outcome (E1). However, none of those exploratory models were significantly better fits than the one that only considered features of the communication context.

Many earlier studies on friendly or cooperative settings have found a link between interpersonal synchrony and rapport (e.g., Hove & Risen, 2009; Hu et al., 2022; Paxton & Dale, 2013b). After failing to find significant effects in our planned and exploratory analyses, we looked back to previous literature and identified another potential effect—the *history* of the dyad's interactions (i.e., the order in which they held the experimental conversations; Paxton & Dale, 2013a). We created an additional exploratory model to account for conversation order in an attempt to shed light as to *why* our expectations were wrong. In this final model, we found that participants' feelings of closeness were complexly affected by the order in which participants had conversations and their movement patterns—but only for arguments in the Laboratory VC condition (as compared with friendly conversations held FTF). This generally supported the trend identified in previous work that arguments play an important role shaping the unfolding pattern of synchrony across multiple conversations (Paxton & Dale, 2013a).

Our findings underscore the complexity of social communication in the dynamical systems tradition (e.g., Richardson & Chemero, 2014). By conceptualizing the dyad as our unit of interest and preserving the temporality of its changes over time (Carello & Moreno, 2005), we are able to better understand the interacting constraints that shape communication patterns and outcomes. Within that tradition, we believe our findings demonstrate the value of considering the system's history and context when studying behavior. It is, of course, difficult to make causal claims between movement patterns and feelings of closeness in the present study, given that synchrony was naturally observed rather than experimentally manipulated. From a dynamical systems perspective, it is just as important to watch the *unfolding* of a process as it is to understand its end-state. As such, it may be better to conceptualize the patterns of movement as part of the process of rapport-building rather than a predictor per se of rapport.

In our preceding subsections, we have already discussed some possible explanations as to why the broader social context might have led to some of the unexpected effects in VC contexts more generally. However, when analyzing closeness, we specifically found differences between FTF and VC Laboratory interactions. One explanation for this difference may be linked to our safety-related methodological choices (as explained in Section 5.1). The laboratory VC condition involved an interesting mixture of interaction medium throughout the session: Participants met FTF at the beginning, but they conducted all their conversations co-located in the lab but video-mediated. This specific interaction history might explain why certain effects were particularly strong when comparing this condition to FTF (all interactions being co-located and face-to-face) and Remote VC (all interactions being separate and video-mediated). This interesting finding could have the most potential to inform work-related questions about effectiveness and feelings

of work-from-home culture and its effects on workers. Future work may investigate whether a mixture of FTF and VC-mediated interactions with a variety of settings could improve feelings of closeness amongst co-workers.

6. Conclusion

Here, we set out to better understand how the amount of visual information, communication medium, and type of conversation affect the ways in which we interact. Generally, we found that interaction dynamics are shaped by each of these factors—though not in the way that we had originally expected, given how deeply social interaction was changed by the pandemic and post-pandemic societal context. In this sense, our study provides strong support for the idea of approaching social interaction as a complex dynamical system (Richardson & Chemero, 2014): Our communication is shaped by how, when, and with what goals we engage with one another.

CRedit authorship contribution statement

Romero: Conceptualization, Methodology, Investigation, Formal analysis, Resources, Data Curation, Writing – Original, review & editing, Supervision, Project Administration.

Paxton: Conceptualization, Methodology, Visualization, Formal analysis, Resources, Data Curation, Writing – Original, review & editing, Supervision, Project Administration.

Declaration of competing interest

The authors have no conflict of interest to declare.

Data availability

We have shared the extracted movement time-series and de-identified participant response data on a restricted-access repository on ICPSR.

Appendix A. Exclusion criteria

All conditions

Dyads who meet any or all of the following criteria will be removed and replaced: (a) failure to complete at least 8 min of conversation for each interaction ($n = 0$); (b) no disagreement during the argumentative interaction ($n = 8$); (c) missing post-interaction questionnaire data for any of the interactions ($n = 2$); (d) withdrawal of consent by one or both participants ($n = 1$); (e) data loss due to technical issues (e.g., recording failures) or experimenter error (e.g., forgetting to turn on microphones; $n = 29$).

FTF condition

In addition to the “All Conditions” criteria, dyads in the FTF condition who meet any or all of the following criteria will also be removed and replaced: (a) participants who fail to follow COVID-19 prevention protocols will not be allowed to complete the study ($n = 0$).

Laboratory VC condition

In addition to the “All Conditions” and “FTF Condition” criteria, dyads in the Laboratory VC condition who meet any or all of the following criteria will also be removed and replaced: (a) those that indicate in their demographic survey that their interactions had “too many interruptions” in their audio or video ($n = 0$).

A.1. Remote VC condition

In addition to the “All Conditions” and the “Laboratory VC Condition” criteria, dyads in the Remote VC condition who meet any or all of the following criteria will also be removed and replaced: (a) failure to follow the setup instructions throughout the experiment ($n = 7$); (b) $>5\%$ of data with camera placement movement from one or both participants; (c) $>5\%$ of data with muted, garbled, or otherwise flawed audio ($n = 0$); (d) recordings that include people who were not the consented participants (e.g., roommates, family members; $n = 2$).

Appendix B. Initial email sent to students in the remote VC condition

Dear XX,

Thank you for signing up to take part in our study entitled “Conversation in the time of COVID-19 - REMOTE.” As was stated in the description of the study, there are certain instructions we will need you to follow. In this study you will be asked to hold conversations with another participant over Zoom. For the duration of the study, you will need to participate with your camera on in three separate conversations. In order for the study to yield the best data, we need you to find a quiet place where you can secure your laptop or desktop such that it does not move in the entirety of the Zoom call and such that no one walks behind you during the conversations. Ideally, you could position yourself with a wall right behind you in a way in which your computer is stationary (on a desk or table, not on your lap). All of these measures will ensure that we capture you and only you during the conversations. We also need you to find a place to take this call without a mask, such that the other participant and researcher can see your full face and while you are on the call we will ask you turn off your self-view on Zoom (in other words, you will only see the video of others during the call and not your own).

We will make sure to repeat these instructions at the beginning of the study session. Thank you for your understanding and cooperation.

If you are unable to follow one or all of the instructions, please reach out to one of us in the research team at our listed email addresses below.

All the best,

Email sent starting July 2022: We added bold text for emphasis and a section about windows given some of the issues we encountered with data collection.

Dear XX,

Thank you for signing up to take part in our study entitled “Conversation in the time of COVID-19 - REMOTE.” As was stated in the description of the study, there are certain instructions we will need you to follow. In this study you will be asked to hold conversations with another participant over Zoom. For the duration of the study, you will need to participate **with your camera on** in three separate conversations. In order for the study to yield the best data, we need you to find a **quiet** place where you can secure your laptop or desktop **such that it does not move in the entirety of the Zoom call** and such that **no one walks behind you during the conversations**. Ideally, you could position yourself with a wall right behind you in a way in which your computer is stationary. Also make sure that there are **no windows behind you**, and if there is one, that it is covered by a curtain or blind. We also need you to find a place to take this call without a mask, such that the other participant and researcher can see your full face. We will make sure to repeat these instructions at the beginning of the study session.

Thank you for your understanding and cooperation. If you are unable to follow one or all of the instructions, please reach out to one of us in the research team at our listed email addresses below.

Looking forward to seeing you (insert date and time).

All the best,

Appendix C. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.actpsy.2023.103992>.

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